

Elias Hossain

PhD Researcher, University of Central Florida

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Research Focus: Trustworthy AI. I design inference-time methods, multi-agent verifiers, and evaluation benchmarks that make large language models more reliable, calibrated, and grounded, with current threads in (1) uncertainty quantification, (2) preference alignment, and (3) trustworthy AI for drug discovery.

EDUCATION

University of Central Florida - Complex Adaptive Systems Lab (CASL)

Ph.D. in Industrial Engineering & Management Systems

- Advisor: Dr. Niloofar Yousefi; Co-Advisor: Dr. Ivan Garibay

Orlando, FL

Aug 2025 - Present

Mississippi State University - PATENT Lab

M.S. in Computer Science (Research); GPA: 3.88/4.00

- Primary Advisor: Dr. Andy D. Perkins; Thesis Advisor: Dr. Shahram Rahimi
- Thesis: A Multi-Stage AI Framework for Topic Discovery in Scientific Abstracts

Mississippi State, MS

Jan 2024 - Jul 2025

RESEARCH EXPERIENCE

UCF Complex Adaptive Systems Lab (CASL)

PhD Researcher & Graduate Research Assistant

- **Trustworthy LLMs:** Designing inference-time methods, multi-agent verifiers, and benchmarks for reliable, calibrated, and grounded large language models.
- **Drug Discovery & AMR:** Building agentic LLM frameworks for interpretable RNA-Seq analysis and foundation-model approaches for antimicrobial resistance prediction.
- **Alignment Research:** Closed-form post-training steering, online preference alignment, and fault-tolerant verification for LLM training pipelines.

Orlando, FL

Aug 2025 - Present

MS State PATENT Lab, Department of Computer Science & Engineering

Graduate Research Assistant

- **Cybersecurity AI:** Contrastive learning framework for anomaly detection in large-scale security logs; fine-tuned domain-specific LLMs for automated threat reasoning.
- **BarkPlug V.2:** FAISS-based RAG pipeline for a university chatbot; evaluated with RAGAS metrics and System Usability Scale (SUS).

Mississippi State, MS

Jan 2024 - Jul 2025

REVE Systems

Software Engineer

- **Speech AI:** Deployed DeepSpeech and Wav2Vec 2.0 (20% lower WER, 15% faster inference); STT/TTS extensions for browser, desktop, and government e-governance services.

Dhaka, Bangladesh

Dec 2022 - Nov 2023

Time Research & Innovation

AI/ML Researcher (Remote)

- Manager: Dr. Shah Siddiqui; Mentor: Dr. Shamsul Masum
- **Medical AI:** Deep learning models (InceptionV3, VGG19, MobileNet) for COVID-19 diagnosis, bone abnormality detection, and diabetes prediction; co-developed a UK telemedicine platform.

Portsmouth, United Kingdom

Oct 2020 - Nov 2022

PUBLICATIONS (FIRST AUTHORED)

Full list on Google Scholar.

1. **Hossain, E.**, Basher, M.J.I., Garibay, I., Garibay, O., & Yousefi, N. “When Policies Cannot Be Retrained: A Unified Closed-Form View of Post-Training Steering in Offline RL,” under review at **NeurIPS 2026** (Main Track).
2. **Hossain, E.**, et al. “MIRAGE-BIO: A Robust Inference-Time Screening System for Biomolecular Candidate Selection under Unreliable Surrogate Evidence,” under review at **NeurIPS 2026** (Main Track).
3. **Hossain, E.**, et al. “BioDivergence: A Leakage-Aware Benchmark for Contextual Biomedical Contradiction Evaluation,” under review at **NeurIPS 2026** (Datasets & Benchmarks Track).
4. **Hossain, E.**, Rahimimovassagh, M., Neupane, S., Basher, M.J.I., Garibay, I., & Yousefi, N. “Fault-Tolerant Preference Alignment via Multi-Agent Verification,” **ICLR 2026 Trustworthy AI Workshop**.
5. **Hossain, E.**, Shoeibi, M., Garibay, I., & Yousefi, N. “BioGen: An Agentic LLM Framework for Drug Discovery via Interpretable RNA-Seq Analysis,” under review at **ISMB 2026**. arXiv:2510.16082
6. **Hossain, E.**, Dipta, S.R., Neupane, S., Rana, R., Shwartz-Ziv, R., & Garibay, I. “UAT-LITE: Inference-Time Uncertainty-Aware Attention for Pretrained Transformers,” under review at **ACL ARR 2026**.
7. **Hossain, E.**, Nipu, M.H., Sheikh, M., Rana, R., Neupane, S., & Yousefi, N. “MedBayes-Lite: Bayesian Uncertainty Quantification for Safe Clinical Decision Support,” under review at **Nature Communications AI & Computing**.

COLLABORATIONS

1. Kansana, M., **Hossain, E.**, Rahimi, S., & Golilarz, N.A. “Surformer v1: Transformer-Based Surface Classification using Tactile and Vision Features,” **Information**, 2025.
2. Golilarz, N.A., **Hossain, E.**, Rahimi, S., & Karimi, H. “A Hybrid Approach for Pattern Recognition and Interpretation in Age-Related False Memory,” **Frontiers in Psychology**, 2025.

3. Neupane, S., **Hossain, E.**, Keith, J., Tripathi, H., Ghiasi, F., et al. "From Questions to Insightful Answers: Building an Informed Chatbot for University Resources," **IEEE FIE 2024**.
4. Kansana, M., Rahimi, K.A., **Hossain, E.**, Dehzangi, I., & Golilarz, N.A. "Edge-Based Learning for Improved Classification under Adversarial Noise," **SEDE 2025** (Springer CCIS, vol. 2720).
5. Golilarz, N.A., **Hossain, E.**, Addeh, A., & Rahimi, K.A. "AI Learning Algorithms: Deep Learning, Hybrid Models, and Large-Scale Model Integration," arXiv:2410.09186, 2024.

PROFESSIONAL SERVICE

Peer Reviewer (2024 - Present), reviewing AI/ML methods for healthcare and biomedical data science.

- **Workshops:** ICLR Trustworthy AI Workshop.
- **Journals:** Scientific Reports (Nature Portfolio), PLOS ONE, Computer Methods in Biomechanics and Biomedical Engineering, Elsevier, Taylor & Francis, Springer.

AWARDS & HONORS

- **Graduate Presentation Fellowship**, UCF, 2025. Selective travel award for presenting research at MIT.
- **Finalist, Dr. Pradeep P. Thevannoor Innovation Awards**, India, 2019.
- **Research Award**, Daffodil International University, 2019. For publication in a reputed indexed journal.
- **Winner, IEEE SS12 Innovation Challenge & Maker Fair**, 2018. Kidnap Prevention Mobile App (IEEE Xplore).
- **Divisional Champion & Global Nominee, NASA Space Apps Challenge**, 2017. Drone for Green Project.

TECHNICAL SKILLS

- **Languages:** Python, R, JavaScript, C#, SQL, Bash.
- **ML/DL:** PyTorch, TensorFlow, Hugging Face Transformers, scikit-learn.
- **LLMs & Alignment:** RLHF, DPO, SFT, Preference Optimization, RAG, Agentic AI, LLM Evaluation.
- **Theory & Modeling:** Reinforcement Learning, Bayesian Modeling, Uncertainty Quantification, Calibration.
- **Infrastructure:** FAISS, Docker, Git, Streamlit, Apache Spark, Azure, GCP.